47202 I.F.E. Problems, Lecture 7 2026

Short questions

1. What is the role of the electrolyte in an electrochemical cell?

2. Which properties must an electrolyte have?

3. How are the anode and the cathode defined?

4. What is the basic difference between thermodynamics and kinetics?

5. Balance the following half-cell reactions$PbSO_4 \rightleftarrows Pb + SO_4^{2-}$ (one of the electrode reactions in a lead-acid battery) (don't look in the text😊)

$Ni(OH)_2 \rightleftarrows NiOOH + H^+$ (the nickel electrode in some batteries)

$O_2 + H_2O \rightleftarrows OH^-$ (the oxygen process in some fuel cells). Note that with OH^-, the system is alkaline and there should be no protons (H^+) in the equation. Protons, if any, would immediately react with OH^- and form water.

$MnO_4^- + H^+ \rightleftarrows MnO_2 + H_2O$ (the cathode process in some batteries). H^+ means an acidic environment and therefore OH^- should not be there.

6. Which combination of reactions from the table with *Standard potentials* (E^o) in the text will give the highest cell voltage? How high is it?

7. Can we make such a battery? Would it be a superior battery?

Calculations:

8. What would the standard cell potential be if a hydrogen/oxygen fuel cell were operated on hydrogen and chlorine gas instead?

9. On fast charging. Consider a battery with the fixed internal ohmic resistance, *R,* of the electrolyte and with catalysts in the electrodes so efficient that the kinetic overpotential is negligible at any practical current. Can such a battery be fast charged without a significant loss of efficiency? Show why/why not?

10. The common alkaline battery is made from manganese and zink and the electrode discharge reactions are:

Anode: $Zn(s) + 2OH^-(aq) \rightleftarrows ZnO(s) + H_2O(l) + 2e^-$

Cathode: $2MnO_2(s) + H_2O(l) + 2e^- \rightleftarrows Mn_2O_3(s) + 2OH^-(aq)$

a) What is the total cell reaction?

b) Calculate the standard cell potential. The standard free energy of formation can be found in the following table.

25 ºC, 1 bar abs.	Zn(s)	ZnO(s)	MnO_2(s)	Mn_2O_3(s)	H_2O(l)
Δg_f^0 (kJ mol^{-1})	0.0	-320.5	-465.1	-881.1	-137.1

11. A high temperature fuel cell is operated on hydrogen and oxygen at 800 ºC. At this temperature, the free energy of formation of hydrogen, oxygen and water are:

800 ºC, 1 bar abs.	H_2	O_2	H_2O
Δg_f (kJ mol^{-1})	0.0 kJ mol^{-1}	0.0 kJ mol^{-1}	-188.5 kJ mol^{-1}

a) What is the standard cell potential at this temperature? The reaction is $H_2 + ½O_2 \rightleftarrows H_2O$

b) What is the reversible cell potential, if the cell is operated on hydrogen and air, each supplied at the pressure 2 bar absolute. Assume that the partial pressure of the produced water is 0.5 bar absolute and that the content of oxygen in air is 20%.

12. A single cell in an industrial electrolyzer is operated for an hour with the current 2000 A. During that period, it produces 72 g hydrogen. The process is

$$H_2O \rightarrow H_2 + ½O_2.$$

Calculate the faradic efficiency of the process. Remember that each mole H_2 requires 2 mole electrons.